RKG Wallah Academy

Class 6 Mathematics — Complete Notes

NCERT based chapter-wise notes with formulas, tables and solved examples.

Note: Ye NCERT-based summary notes hain. Diagrams NCERT se apni copy mein banao aur exam se pehle apni book se ek baar mila lo.

1. Knowing Our Numbers

• Place value: ones, tens, hundreds, thousands, ten-thousands, lakhs, ten-lakhs, crores (Indian system). International: ones, thousands, millions, billions.

• 1 lakh = 100 thousand = 10^5; 1 crore = 100 lakh = 10^7; 1 million = 10 lakh; 1 billion = 100 crore.

• Greatest 4-digit number 9999; smallest 1000. Greatest 5-digit 99999; smallest 10000.

• Estimation: rounding off to nearest 10/100/1000 (5 or more → round up).

• Roman numerals: I=1, V=5, X=10, L=50, C=100, D=500, M=1000. A symbol repeats at most 3 times; a smaller before a larger is subtracted (IX = 9, XL = 40).

2. Whole Numbers

• Natural numbers: 1, 2, 3, ... Whole numbers: 0, 1, 2, 3, ... (natural numbers + 0). Every natural number is whole; 0 is the smallest whole number.

• Properties: closure under addition and multiplication; commutative (a+b = b+a, a×b = b×a); associative; distributive: a×(b+c) = a×b + a×c.

• Additive identity = 0; multiplicative identity = 1. Division by zero is not defined.

• Successor = n + 1; predecessor = n − 1.

3. Playing with Numbers

Factors and multiples

• Factor divides the number exactly; every number has 1 and itself as factors. Multiples of n: n, 2n, 3n, ...

• Prime number: exactly two factors (2, 3, 5, 7, 11...). 2 is the only even prime. Composite: more than two factors. 1 is neither prime nor composite.

• Co-prime numbers: HCF is 1 (e.g., 8 and 15). Twin primes: differ by 2 (11, 13).

Divisible by	Test
2	Last digit 0, 2, 4, 6, 8
3	Sum of digits divisible by 3
4	Last two digits divisible by 4
5	Last digit 0 or 5
6	Divisible by 2 and 3
8	Last three digits divisible by 8
9	Sum of digits divisible by 9
10	Last digit 0
11	Difference of sum of digits at odd and even places is 0 or multiple of 11

• **Prime factorisation:** writing a number as product of primes (e.g. 60 = 2×2×3×5).

• **HCF** (Highest Common Factor): greatest number dividing all the given numbers. **LCM**: smallest number that is a multiple of all.

• For two numbers: HCF × LCM = product of the numbers.

Q. Find HCF and LCM of 12 and 18.

Sol. $12 = 2^2 \times 3$; $18 = 2 \times 3^2$. HCF = $2 \times 3 = \mathbf{6}$; LCM = $2^2 \times 3^2 = \mathbf{36}$. Check: $6 \times 36 = 216 = 12 \times 18$.

4. Basic Geometrical Ideas

• **Point** (no size), **line segment** (two end points, measurable), **line** (extends both ways), **ray** (one end point).

• Parallel lines never meet; intersecting lines meet at one point. Curves: open/closed, simple.

• Polygon: closed figure of line segments (triangle 3 sides, quadrilateral 4, pentagon 5, hexagon 6).

• Circle: centre, radius, diameter (= 2 × radius), chord, arc, sector, segment, semicircle.

5. Understanding Elementary Shapes

Angle	Measure
Acute	Less than 90°
Right	Exactly 90°
Obtuse	Between 90° and 180°
Straight	180°
Reflex	Between 180° and 360°
Complete	360°

• Complementary angles add up to 90°; supplementary angles add up to 180°.

• Triangles by sides: scalene, isosceles (two equal sides), equilateral (all three). By angles: acute, right, obtuse. Sum of angles of a triangle = 180°.

• Quadrilaterals: square, rectangle, parallelogram, rhombus, trapezium. Sum of angles = 360°.

• 3-D shapes: cube (6 faces, 12 edges, 8 vertices), cuboid, cylinder, cone, sphere, prism, pyramid.

6. Integers

• Integers: …, −3, −2, −1, 0, 1, 2, 3, … Positive on right of zero, negative on left on a number line.

• Adding a negative = moving left. Subtraction: a − b = a + (−b).

• Absolute value |−5| = 5. Opposite of 7 is −7; sum of an integer and its opposite is 0.

• Of two integers, the one on the right is greater.

7. Fractions

• Fraction = numerator/denominator. Proper (num < den), improper (num ≥ den), mixed (3 ½), equivalent fractions (multiply/divide num and den by same number).

• Like fractions: same denominators. To add/subtract unlike fractions, make denominators equal (use LCM).

• Compare: convert to same denominator or cross multiply. Simplest form: divide by HCF.

• Multiplication: (a/b)×(c/d) = ac/bd. Division: multiply by reciprocal.

Q. Add 2/3 and 3/4.

Sol. LCM of 3 and 4 = 12. 8/12 + 9/12 = **17/12 = 1 5/12**.

8. Decimals

• Decimal places: tenths (0.1), hundredths (0.01), thousandths (0.001). 0.5 = 5/10 = 1/2.

• To compare, compare whole parts and then digit by digit from the left.

• Adding/subtracting: align decimal points. Multiplying by 10, 100, 1000 shifts the point to the right.

• Money: 100 paise = ₹1; length: 1 cm = 10 mm, 1 m = 100 cm, 1 km = 1000 m; weight: 1 kg = 1000 g.

9. Data Handling

- Data is collected as **raw data**; **tally marks** (groups of five) make a frequency table.
- **Pictograph:** uses pictures where 1 symbol = given number of items.
- **Bar graph:** bars of equal width and equal spacing; height shows the quantity.

10. Mensuration

Perimeter of rectangle = 2(l + b) Square = 4a Equilateral triangle = 3a

Area of rectangle = l × b Area of square = a × a (Area unit: square units, e.g., cm²)

Q. A rectangular field is 30 m long and 20 m wide. Find the perimeter and area.

Sol. Perimeter = 2(30+20) = **100 m**; Area = 30×20 = **600 m²**.

11. Algebra

• Variable: a letter that can take different values. Expression: made from variables and numbers (like 3x + 5).

• Rules from patterns: perimeter of square = 4l; of equilateral triangle = 3l; number of diagonals etc.

• Equation: statement that two expressions are equal. Solution is the value of the variable that makes it true. Solve by transposing: x + 5 = 12 → x = 7; 3x = 15 → x = 5.

12. Ratio and Proportion

• Ratio compares two quantities of the same kind in the same units: a : b = a/b. Simplest form by dividing by common factor.

• Proportion: equality of two ratios, a : b :: c : d, i.e., a×d = b×c.

• Unitary method: find value of one unit first, then of required units.

Q. If 6 pens cost ₹90, what is the cost of 10 pens?

Sol. Cost of 1 pen = 90/6 = ₹15. Cost of 10 pens = 15×10 = **₹150**.

13. Symmetry and Practical Geometry

• A figure has line symmetry if it can be folded along a line so that two halves coincide exactly. Square: 4 lines; rectangle: 2; equilateral triangle: 3; circle: infinite; letter A, M, T have one line.

• Practical geometry: using ruler, compass and protractor to draw a circle, line segment, perpendicular bisector, angle of given measure, angle bisector.